

Customer Configuration Guide

(prod-v0.2.6)

This document explains the different knobs available for configuration of Datology's curation product.

Wrapped Stew 001

Required Arguments

`source_documents: list[IngestionSource]`

The datasets to be curated. Each `IngestionSource` has:

- `name` (str, required): Identifier for this source (e.g., "dclm", "fineweb"). Used to reference the source in other arguments like `web_dataset_names`.
- `source_location` (str, required): Path to the raw source data (e.g., "s3://bucket/path/").
- `ingested_location` (str, optional): Path to already-ingested data. If set, ingestion is skipped for this source, and none of the below arguments are required/used.
- `override_args` (dict, required): Custom ingestion arguments. Supported keys:
 - `text_column` (str, required): Column name of the text column in the data source
 - `file_type`: (str, optional): currently only allowed "parquet" - other formats incoming.
 - `metadata_column_mapping` (dict, optional): Mapping from additional fields inside of the input to fieldnames to store as metadata for retrieval later. When supplied, no columns other than `text_column` and columns here end up in the ingested data. Special internal-use fields are "external_id", "url", so you may do something like {"your_id_name": "external_id", "url": "url", "other_metadata_field_we_want_to_keep": "field_name"}. See `export_args["metadata_fields"]` for more information on retrieving these in your final output.
- `max_row_size` (int, optional): if set, discards rows that are greater than `max_row_size` bytes. For text files, individual rows larger than 1MB may currently have issues with the tokenizer flow in logographic languages.

`target_documents: list[IngestionSource]`

Target/evaluation datasets used for distribution-matching curation strategies. Same structure as `source_documents`.

`target_output_tokens: int`

The total number of output tokens to produce across all domains. This is split across all domains and sources according to the release's `mixture_proportions`.

`export_format: str`

Output format for the curated data. Supported values (only "parquet" for now):

- "parquet": Apache Parquet with field selection.

`export_args: optional[dict]`

Optional additional export args. Currently supported:

- `metadata_fields` (list of metadata field names to include). This can be any of the metadata fields you ingested by name (with their datology-local name, the value side of the metadata mapping dictionary), and these additional special fields:
 - `schema`: includes the origin source name from `source_documents` for this row
 - `synthetic_model`: provides the HuggingFace path to the model used to produce this example. Null if the row did not use any synthetic models during processing.

`web_dataset_names: list[str]`

Names of sources (from `source_documents`) to use for web curation. These are processed through quality filtering, distribution-matching, and synthetic rephrasing.

`web_target_names: list[str]`

Names of targets (from `target_documents`) to use as sources for distribution fitting. These targets are never directly used as selectable documents in our curation flows, nor are they used as inputs to or observed by our rephrasing models.

`web_decontamination_target_names: list[str]`

Names of targets (from `target_documents`) to use as sources for distribution fitting. These targets are never directly used as selectable documents in our curation flows, nor are they used as inputs to or observed by our rephrasing models.

Optional Arguments

`output_datasets_path`: optional[str]

If set, the final curated datasets are promoted/moved to this location. If None, datasets remain at their pipeline-generated location.

`tokenizer_name`: str

Tokenizer used for token counting. Default: "EleutherAI/gpt-neox-20b".

`code_dataset_names`: list[str] | None

Names of sources to use for code curation. If None or empty, code curation is skipped entirely.

`math_dataset_names`: list[str] | None

Names of sources to use for math curation. If None or empty, math curation is skipped.

`multilingual_dataset_names`: dict[str, list[str]] | None

Mapping of language names to source names for multilingual curation. If None or empty, multilingual curation is skipped.

Supported languages (can use either the full lowercase names or the 2-letter language codes):

(arabic, ar), (bengali, bn), (dutch, nl), (filipino, tl), (french, fr), (german, de), (hindi, hi), (indonesian, id), (italian, it), (japanese, ja), (korean, ko), (mandarin, zh), (marathi, mr), (portuguese, pt), (russian, ru), (spanish, es), (tamil, ta), (telugu, te), (urdu, ur), (vietnamese, vi)

Example:

```
multilingual_dataset_names={
    "es": ["fineweb"],
    "fr": ["fineweb", "dclm"],
}
```

`multilingual_target_names`: list[str] | None

Target datasets for multilingual mining. Defaults to `web_target_names` if not provided.

`code_included_languages: list[str] | None`

Specific programming languages to include in code curation. If None, no inclusion filter is applied.

`code_excluded_languages: list[str] | None`

Specific programming languages to exclude from code curation. If None, no exclusion filter is applied.

Model Selection

Each domain allows customizing which models are used for rephrasing or translation:

- **web_rephrase_models (Web rephrasing)**
 - Default: PHI4, MISTRAL_7B_V3, QWEN3_8B, OLMO2_7B
- **code_beyondweb_rephrase_models (Code BeyondWeb rephrasing)**
 - Default: PHI4, MISTRAL_7B_V3, QWEN3_8B, GPT_OSS_20B
- **code_synthetic_rephrase_models (Code Synthetic rephrasing)**
 - Default: QWEN_2_5_CODER_7B_INSTRUCT, MISTRAL_7B_V3
- **math_rephrase_models (Math rephrasing)**
 - Default: QWEN3_235B_A22B_INST_2507_FP8, GPT_OSS_120B
- **multilingual_translation_models (Multilingual translation)**
 - Default: QWEN3_8B, PHI4

All accept `list[ModelName] | None`. Set to None to use defaults. Available model names are:

"gpt-oss-20b", "olmo2-7b", "olmo3-7b-instruct", "phi4", "mistral-7b-v3",
"ministral-3-8b-instruct", "qwen2.5-coder-7b-instruct", "qwen3-8b", "qwen3-5-9b",
"gpt-oss-120b" (h100 only), "qwen3-235b-a22b-inst-2507-fp8" (h100 only)

`postprocess_synthetic_code: bool`

If set to True, postprocesses synthetic code via heuristic filters and cleaning methods. Defaults to False. Requires that coding datasets have specified a `language` column

`global_dedup_max_duplicates_per_key: int | None`

The maximum number of duplicate documents allowed per deduplication key during global dedup. Must be ≥ 0 if set. If None, global dedup is not run. **Note:** if using this argument, the curation product may return fewer tokens than desired due to deduplication.

`format_markdown: bool`

Whether to apply standard formatting to markdown documents to make them more easily processed (currently applies compaction to all markdown tables). Defaults **False**

`filter_urls: list[str] | None = None`

Blocklist of keywords or regexes (in the format `"re:<target_regex>"`) for URLs you'd like filtered out of the inputs. Requires supplying the `url` metadata column in your ingestion source.

`min_token_length: int | None`

Set a minimum document length for inputs - input rows shorter than this will be removed before other processing.

`override_args: dict | None`

Advanced overrides for fine-grained control of the underlying pipeline. Structured as:

```
override_args={
  "cluster_config_args": {
    "gpu_num_partitions": <int>,    # Number of GPU partitions for rephrase jobs
    "rephrase_num_workers": <int>,  # Number of Ray workers for rephrasing
    "rephrase_machine_type": <str>,  # Machine type for rephrase workers
  },
}
```

cluster_config_args: Controls Ray cluster sizing and GPU partitioning for rephrase/translation jobs. Helps scale job speed across differently sized jobs.

Register Dataset

The register dataset is launched with the following:

```
Shell
datology add-dataset \
  --run-name desired-dataset-name \
  --output-dataset-path desired-output-dataset-path \
  --extra-curation-args args-file.json
```

The “extra-curation-args” should point to a JSON with the following args:

- **name** (str, required): Identifier for this source (e.g., "dclm", "fineweb"). Used to reference the source in other arguments like `web_dataset_names`.
- **source_location** (str, required): Path to the raw source data (e.g., "s3://bucket/path/").
- **override_args** (dict, required): Custom ingestion arguments. Supported keys:
 - **text_column** (str, required): Column name of the text column in the data source

- `file_type`: (str, optional): currently only allowed "parquet" - other formats incoming.
- `metadata_column_mapping` (dict, optional): Mapping from additional fields inside of the input to fieldnames to store as metadata for retrieval later. Special internal-use fields are "external_id", "url", so you may do something like {"your_id_name": "external_id", "url": "url", "other_metadata_field_we_want_to_keep": "field_name"}. See `export_args["metadata_fields"]` for more information on retrieving these in your final output.
- `max_row_size` (int, optional): if set, discards rows that are greater than `max_row_size` bytes. For text files, individual rows larger than 1MB may currently have issues with the tokenizer flow in logographic languages.

Stew 2 - Web + Multilingual

We're in an *early* public release of this functionality, and are still working on cleaning up the configuration surface (after which we can release a Wrapped Stew 2). If you'd like to use this directly, we can provide a corresponding configuration file based on what you would use to configure Wrapped Stew 1.